JMIR Publications
The Leading eHealth Publisher

Assessment of public awareness, risk perception and mental health burden during the first 3 months of the COVID-19 epidemic in China: a social media surveillance

Zhiyuan Hou, Xinyu Zhou, Hao Jiang, Fanxing Du, Heidi Larson, Leesa Lin

Submitted to: Journal of Medical Internet Research
on: May 20, 2020

Disclaimer: © The authors. All rights reserved. This is a privileged document currently under peer-review/community review. Authors have provided JMIR Publications with an exclusive license to publish this preprint on it's website for review purposes only. While the final peer-reviewed paper may be licensed under a CC BY license on publication, at this stage authors and publisher expressively prohibit redistribution of this draft paper other than for review purposes.

Table of Contents

Original Manuscript 4
Supplementary Files 12
Figures 13
Figure 1 14
Figure 2 15
Figure 3 16
Multimedia Appendixes 17
Multimedia Appendix 1 18
Multimedia Appendix 2 18

Preprint JMIR Publications

Assessment of public awareness, risk perception and mental health burden during the first 3 months of the COVID-19 epidemic in China: a social media surveillance

Zhiyuan Hou[1] PhD; Xinyu Zhou[1]; Hao Jiang[1]; Fanxing Du[1]; Heidi Larson[2] PhD; Leesa Lin[3] PhD

[1]School of Public Health, NHC Key Laboratory of Health Technology Assessment, Fudan University Shanghai CN
[2]Department of Infectious Disease Epidemiology, London School of Hygiene & Tropical Medicine London GB
[3]Department of Infectious Disease Epidemiology, London School of Hygiene & Tropical Medicine London CN

Corresponding Author:
Zhiyuan Hou PhD
School of Public Health, NHC Key Laboratory of Health Technology Assessment, Fudan University
130 Dong'an Road
Shanghai
CN

Abstract

Background: The COVID-19 epidemic and subsequent containment measures may inflate public awareness, risk perception and mental health burden.

Objective: To assess the public awareness, risk perception and mental health burden in real time during the COVID-19 epidemic in China.

Methods: We collected data from Sina Weibo, the most popular social media platforms in China, from December 2019 to March 2020. The daily number of Weibo posts with keywords related to COVID-19 were presented with daily data on confirmed COVID-19 cases. All Weibo posts were analysed by the Linguistic Inquiry and Word Count text analysis program to assess the public risk perception and mental health burden, measured by anxiety, depression, and anger.

Results: A total of 4,992,731 Weibo posts were collected. The first coronavirus death triggered public perceived risk of an imminent epidemic. Delayed release of epidemic information contributed to mental health burden. Mental health burden evolved with the uncertainty and consequent anxiety about the coronavirus epidemic. Public anxiety started to decline after confirmation of human-to-human transmission, yet depression tied to a bleak future in light of the epidemic and anger at the late dissemination of relevant information increased until the implementation of containment measures.

Conclusions: Early release of information on emerging infectious diseases and early implementation of containment measures might effectively manage public awareness and risk perception and the consequential mental health burden. Social media surveillance should be fully incorporated into epidemic preparedness and response systems.

(JMIR Preprints 20/05/2020:20498)
DOI: https://doi.org/10.2196/preprints.20498

Preprint Settings

1) Would you like to publish your submitted manuscript as preprint?

✔ **Please make my preprint PDF available to anyone at any time (recommended).**
Please make my preprint PDF available only to logged-in users; I understand that my title and abstract will remain visible to all users.
Only make the preprint title and abstract visible.
No, I do not wish to publish my submitted manuscript as a preprint.

2) If accepted for publication in a JMIR journal, would you like the PDF to be visible to the public?

✔ **Yes, please make my accepted manuscript PDF available to anyone at any time (Recommended).**
Yes, but please make my accepted manuscript PDF available only to logged-in users; I understand that the title and abstract will remain v
Yes, but only make the title and abstract visible (see Important note, above). I understand that if I later pay to participate in <a href="http

Original Manuscript

Preprint JMIR Publications

Short paper

Zhiyuan Hou, PhD,[1] Xinyu Zhou, Undergraduate,[1] Hao Jiang, Undergraduate,[1] Fanxing Du, MSc,[1] Heidi Larson[2,3], Leesa Lin, PhD[2]

Affiliations:

1. School of Public Health, NHC Key Laboratory of Health Technology Assessment, Fudan University, Shanghai, China
2. Department of Infectious Disease Epidemiology, London School of Hygiene & Tropical Medicine, UK
3. Department of Health Metrics Sciences, University of Washington, Seattle, the U.S.A

Corresponding author: Zhiyuan Hou. School of Public Health, NHC Key Laboratory of Health Technology Assessment, Fudan University, Shanghai, China.
Address: 130 Dong'an Road, Shanghai 200032, China.
Email: zyhou@fudan.edu.cn

Assessment of public awareness, risk perception and mental health burden during the first 3 months of the COVID-19 epidemic in China: a social media surveillance

Abstract

Background: The COVID-19 epidemic and subsequent containment measures may inflate public awareness, risk perception and mental health burden.

Objectives: To assess the public awareness, risk perception and mental health burden in real time during the COVID-19 epidemic in China.

Methods: We collected data from Sina Weibo, the most popular social media platforms in China, from December 2019 to March 2020. The daily number of Weibo posts with keywords related to COVID-19 were presented with daily data on confirmed COVID-19 cases. All Weibo posts were analysed by the Linguistic Inquiry and Word Count text analysis program to assess the public risk perception and mental health burden, measured by anxiety, depression, and anger.

Results: A total of 4,992,731 Weibo posts were collected. The first coronavirus death triggered public perceived risk of an imminent epidemic. Delayed release of epidemic information contributed to mental health burden. Mental health burden evolved with the uncertainty and consequent anxiety about the coronavirus epidemic. Public anxiety started to decline after confirmation of human-to-human transmission, yet depression tied to a bleak future in light of the epidemic and anger at the late dissemination of relevant information increased until the implementation of containment measures.

Conclusions: Early release of information on emerging infectious diseases and early implementation of containment measures might effectively manage public awareness and risk perception and the consequential mental health burden. Social media surveillance should be fully incorporated into epidemic preparedness and response systems.

Keywords: COVID-19, social media, public response, awareness, risk, mental health.

Introduction

The novel coronavirus disease 2019 (COVID-19) emerged in early December 2019 in Wuhan city, Hubei province, and quickly spread across China and worldwide.[1,2] From 23rd January 2020, the Chinese government put Wuhan and several nearby cities under quarantine and nationally implemented containment measures to control community transmission. It was declared as a global pandemic by the WHO on 11th March.[3] The COVID-19 epidemic and subsequent containment measures may inflate public awareness, risk perception and mental health burden.[4,5] Governments should attempt to understand mental health burden during a public health emergency to inform their outreach strategies.

Social media is increasingly recognized as a platform of understanding and responding to public awareness and sentiment.[6-9] In contrast with traditional surveys, social media surveillance can systematically monitor the public responses to the epidemic in real-time and is less likely to be affected by recall bias. Despite these benefits, social media surveillance has been under-explored during the epidemic, with only few studies using Twitter and Facebook on A(H1N1) and Ebola.[10,11] Using social media surveillance data, this study aimed to monitor and analyse the public awareness, risk perception and mental health burden in real time during the first three months of the COVID-19 epidemic in China.

Methods

Similar to Twitter, Sina Weibo (aka “microblog”) is a popular social media platform in China which has more than 500 million registered users who express their views and sentiment real-time in public posts. We collected all COVID-19 related Weibo posts from December 2019 to March 2020, and then excluded posts that were not expressing public sentiment about COVID-19 such as news, advertisements, etc. A list of Weibo keywords about COVID-19 was manually scanned from the first five hundred posts and tested for sensitivity and specificity before being used. All posts gathered were sorted by date and user location (i.e. being either in or out of Hubei province). We referred to Weibo's “Hot Search” ranking to identify critical events and/or topics about COVID-19 that incited the most discussions each day (Appendix Table 1).

The daily number of Weibo posts with keywords related to COVID-19 were presented with daily data on confirmed COVID-19 cases. All Weibo posts were analysed by the Linguistic Inquiry and Word Count (LIWC) text analysis program,[12,13] previously validated in China for Weibo posts[14,15], to assess the public risk perception and mental health burden, measured by anxiety, depression, and anger (Appendix Table 2). Examples of keywords and Weibo posts assessing risk perception and mental health burden are presented in Appendix Table 2. We further conducted regression models to estimate the provincial differences in risk perception and mental health burden across different periods.

Results

Trends of public awareness about COVID-19

A total of 4,992,731 Weibo posts were collected from December 2019 to March 2020, including those in Hubei province (242,800, 4.9%) and outside of Hubei (4,749,931, 95.1%). Figure 1 presented the daily number of posts were presented with confirmed COVID-19 cases in the country. There were no COVID-19 related posts throughout the entirety of December 2019. Post counts jumped to 13,640 on 31st December 2019, when Wuhan Health Commission first announced the COVID-19 outbreak. Post counts increased sharply from 20th January 2020, when the National Health Commission of China confirmed human-to-human transmission, and peaked at 185,104 posts on 23rd January, when Wuhan city started to implement a series of containment measures. Most posts discussed the epidemic and Wuhan city quarantine.

Trends of public risk perception about COVID-19

Figure 2 presented the daily proportions of Weibo posts expressing risk perception, with strong fluctuations before 20th January caused by a small data size (many days Weibo posts <1000). On 31st December 2019, 32.0% of Weibo posts in Hubei province started to recognize the risk of this disease. On 3rd and 5th January, the Wuhan Health Commission repeatedly stated “there was no obvious evidence of human-to-human transmission”, and along with these announcements, public risk perception showed steady reduction for several days. On 11-13 January when the first COVID-19 death in Wuhan and the first confirmed case outside of China were reported, the proportion of posts expressing perceived risk reached a peak of around 75% across China. Public risk perception was high between the first death reported on 11th January and the confirmation of human-to-human transmission on 20th January, and had stabilised with two exceptions: the day after the WHO declared the COVID-19 outbreak a public health emergency of international concern (PHEIC), and the day when the WHO increased the COVID-19 risk to very high.

Trends of public mental health burden about COVID-19: anxiety, depression, and anger

During the first three months of the COVID-19 epidemic, the total proportions of Weibo posts expressing mental health burden, measured by anxiety, depression, and anger, were 11.7%, 6.3%, and 8.5%, respectively (Table 1). As seen in Figure 3, all three indicators responded immediately to four critical events during the COVID-19 epidemic, manifested

within 24 hours after the triggering events took place. Reports of the first confirmed case in China (on 31st December) and then two weeks later abroad (13th January) stirred more anxiety, depression, and anger outside of Hubei province due to concern about the imported cases from Hubei (Hubei vs outside of Hubei: 23.3% vs 29.0% for anxiety, 10.0% vs 16.2% for depression, and 8.3% vs 18.0% for anger on 13th January, respectively). On 6th February, the news of Wenliang Li, the doctor who first warned of the disease risk on social media, died from COVID-19, weighed heavily on public mental health across the country, especially in Hubei (around 23% vs 14% outside of Hubei for each indicator). The WHO declaration of increasing the risk of COVID-19 to very high, triggered another peak of all three indicators on 29th February.

It is worth noting that the confirmation of human-to-human transmission on 20th January marked the beginning of a steady decline in public anxiety; during the same period, public depression and anger increased, but started to decline after the nationwide implementation of containment measures on 23-25 January. The regression showed there is no evidence of difference between residents in Hubei and outside of Hubei in their risk perception and mental health burden before 20th January (Table 1). However, Hubei residents were significantly more likely to have higher perceived risk and mental health burden than those outside of Hubei after the confirmation of human-to-human transmission (20th January).

Table 1: Proportions of Weibo posts expressing risk perception and mental health burden, and their association with location in the different periods of COVID-19 epidemic

	Risk perception	Anxiety	Depression	Anger
Proportion of related posts (%)	36.2	11.7	6.3	8.5
Subsample between 31 December 2019 and 19 January 2020				
Proportion of related posts (%)	40.1	15.5	5.2	7.6
Location #				
Outside of Hubei	0(ref)	0(ref)	0(ref)	0(ref)
Hubei province	0.0074	0.0271	-0.0009	-0.0150
	(0.0409)	(0.0244)	(0.0007)	(0.0150)
Subsample between 20 January and 9 February, 2020				
Proportion of related posts (%)	36.8	12.3	6.8	9.1
Location #				
Outside of Hubei	0(ref)	0(ref)	0(ref)	0(ref)
Hubei province	0.0524**	0.0311**	0.0241**	0.0170*
	(0.0248)	(0.0130)	(0.0100)	(0.0099)
Subsample between 10 February and 30 March, 2020				
Proportion of related posts (%)	34.4	9.9	6.5	8.7
Location #				
Outside of Hubei	0(ref)	0(ref)	0(ref)	0(ref)
Hubei province	0.0270***	0.0123**	0.0177***	0.0182***
	(0.0081)	(0.0055)	(0.0023)	(0.0031)

*# Coefficient and standard errors (in parentheses) were obtained from linear regressions. Significance level: *** p<0.01, ** p<0.05, * p<0.1.*

Discussion

Through monitoring and analysing social media posts, we found that the first coronavirus death triggered the public awareness and perceived risk of an imminent epidemic, and delayed release of information about the coronavirus epidemic contributed to public mental health burden, measured by anxiety, depression, and anger. The lack of timely and effective risk communication in the early epidemic stage would prevent the public from realizing the risk of the disease, and inaccurate information spread on social media would arouse the public anxiety. Public anxiety started to decline after the confirmation of human-to-human transmission, yet both depression and anger increased until the implementation of containment measures. The mental health burden evolved from uncertainty and consequent anxiety about whether or not there was human-to-human transmission, to anger at the late dissemination of relevant information and depression tied to a bleak future in light of the epidemic.

This study showed that despite uncertainty and incomplete information about the risk, the early release of information on emerging infectious diseases and early implementation of containment measures might effectively manage public awareness and risk perception and the consequential mental health burden, which were highly correlated with epidemic news events, government announcements, and the implementation of containment measures. Acknowledging the knowns and unknowns about an emerging health threat and taking proactive mitigation measures helps to ease public mental health burden. As the containment measures continued, the risk perception and mental health burden kept steady. Multifaced strategies are needed to address the mental health burden among the public during the epidemic.

Compared with traditional surveys, social media surveillance provides the real-time, longitudinal, and dynamic data.[6,7] It can systematically monitor and respond to public awareness and mental health burden as they emerge and evolve in real-time; as such, it should be fully incorporated into the epidemic response systems.[16,17] There are several limitations. First, this is not a diagnosis of mental health status. Some users may not post their response and mental health status in social media, leading to underestimation. Trends analysis may reduce this estimation bias. Second, LIWC detects mental health burden using the frequency of mental health related words instead of screening and assessment tools, which may lead to misclassification. We used mental health scales in guiding keywords selection to maximise accuracy. Third, there is an inherent bias in social media studies where users might present themselves differently online and/or represent a skewed population towards the young.

Preprint JMIR Publications

Figure legend

Figure 1: Daily number of COVID-19 related Weibo posts and confirmed COVID-19 cases in China

Notes: Key events related to COVID-19 epidemic were identified on Weibo "Hot Search" ranking. PHER: public health emergency responses; PHEIC: public health emergency of international concern.

Figure 2: Daily proportions of Weibo posts expressing risk perception

Notes: Key events related to COVID-19 epidemic were identified on Weibo "Hot Search" ranking. PHER: public health emergency responses; PHEIC: public health emergency of international concern.

Figure 3: Daily proportions of Weibo posts expressing anxiety, depression and anger

Notes: Key events related to COVID-19 epidemic were identified on Weibo "Hot Search" ranking. PHER: public health emergency responses; PHEIC: public health emergency of international concern.

Multimedia Appendix

Appendix Table 1: Ranking of key events in Weibo "Hot Search" during the COVID-19 epidemic

Appendix Table 2: Examples of keywords and Weibo posts assessing risk perception and mental health burden by the Linguistic Inquiry and Word Count program

Preprint JMIR Publications

Acknowledgements
ZH and LL conceptualized the study design. XZ, HJ, FD collected and analysed data. ZH, LL, HJ, HL interpreted results and wrote the manuscript. All authors approved the final draft of the manuscript. Z.H. acknowledges financial support from the National Natural Science Foundation of China (No. 71874034). The funders had no role in the study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests
We declare no competing interests.

Abbreviations
COVID-19: coronavirus disease 2019
WHO: World Health Organization
PHEIC: public health emergency of international concern
LIWC: the Linguistic Inquiry and Word Count

Preprint
JMIR Publications

References

1. Huang C, Wang Y, Li X, et al. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. Lancet 2020; **395**(10223): 497-506.
2. Chen N, Zhou M, Dong X, et al. Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study. Lancet 2020; **395**(10223): 507-513.
3. World Health Organisation. WHO Director-General's opening remarks at the media briefing on COVID-19 - 11 March 2020. (accessed March 11, 2020). https://www.who.int/dg/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19---11-march-2020
4. Lin L, Jung M, McCloud RF, Viswanath K. Media use and communication inequalities in a public health emergency: a case study of 2009-2010 pandemic influenza A virus subtype H1N1. *Public health reports (Washington, DC : 1974)* 2014; 129 Suppl 4(Suppl 4):49-60.
5. John Sumo et al. Risk and Outbreak Communication: Lessons from Taiwan's Experiences in the Post-SARS Era. *Health Secur* 2017; **15**(2): 165-169.
6. Gabriel J Milinovich, Gail M Williams, Archie C A Clements, Wenbiao Hu. Internet-based surveillance systems for monitoring emerging infectious diseases. *Lancet Infect Dis* 2014; **14**(2): 160-8.
7. Tang L, Bie B, Park SE, Zhi D. Social media and outbreaks of emerging infectious diseases: A systematic review of literature. *Am J Infect Control* 2018; **46**(9): 962-972.
8 Fung IC, Duke CH, Finch KC, Snook KR, Tseng PL, Hernandez AC, Gambhir M, Fu KW, Tse ZTH. Ebola virus disease and social media: A systematic review. *Am J Infect Control* 2016; **44**(12): 1660-1671.
9. The Lancet. The medium and the message of Ebola. *Lancet* 2014; **384**: 1641.
10. Fung IC, Tse ZT, Cheung CN, Miu AS, Fu KW. Ebola and the social media. *Lancet* 2014; **384**(9961): 2207.
11. Signorini A, Segre AM, Polgreen PM (2011). The Use of Twitter to Track Levels of Disease Activity and Public Concern in the U.S. during the Influenza A H1N1 Pandemic. *PLOS ONE* 2011; **6**(5): e19467.
12. Pennebaker J, Boyd R, Jordan K, Blackburn K. The development and psychometric properties of LIWC2015, Austin, TX: University of Texas at Austin;2015.
13. Tausczik Y, Pennebaker J. The psychological meaning of words: LIWC and computerized text analysis methods. *J Lang Soc Psychol* 2010; **29**:24–54.
14. □□□, Chung C K , Hui N , et al. □□□□□□□□□□□□□□□□□□□□. □□□□□□ 2012, **54**(2):185-201.
15. Zhao N, Jiao D, Bai S, et al. Evaluating the Validity of Simplified Chinese Version of LIWC in Detecting Psychological Expressions in Short Texts on Social Network Services. *PLOS ONE* 2016, **11**(6):1-15.
16. The Lancet. COVID-19: fighting panic with information. *Lancet* 2020; **395**(10224): 537.
17. Depoux A, Martin S, Karafillakis E, Bsd RP, Wilder-Smith A, Larson H. The pandemic of social media panic travels faster than the COVID-19 outbreak. *J Travel Med*. 2020. pii: taaa031. doi: 10.1093/jtm/taaa031.

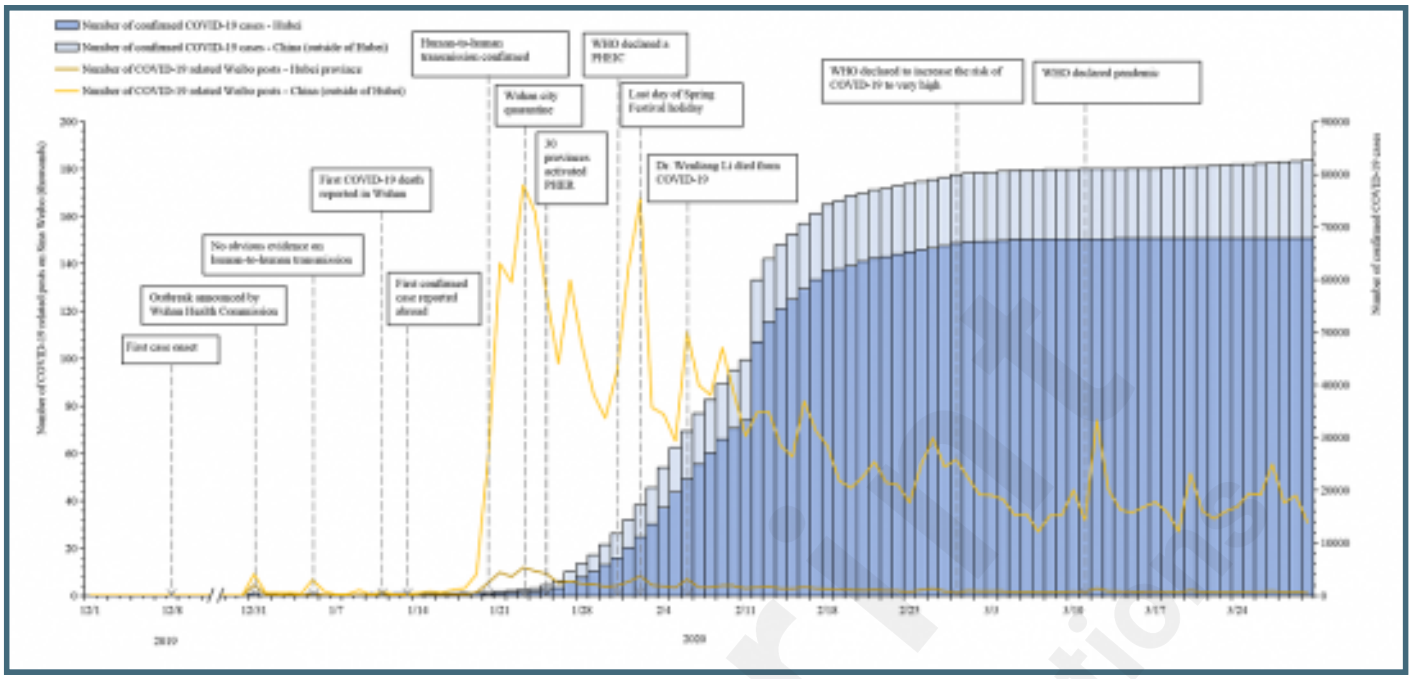
Supplementary Files

Preprint JMIR Publications

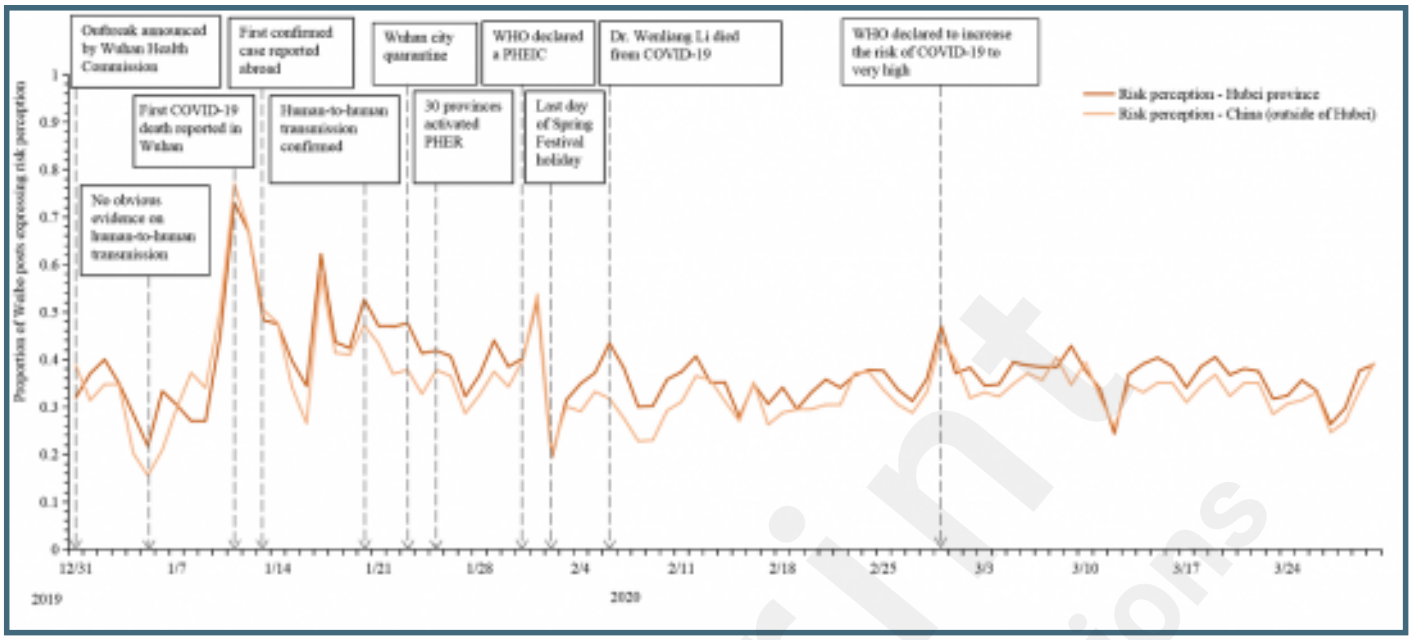
Figures

Preprint JMIR Publications

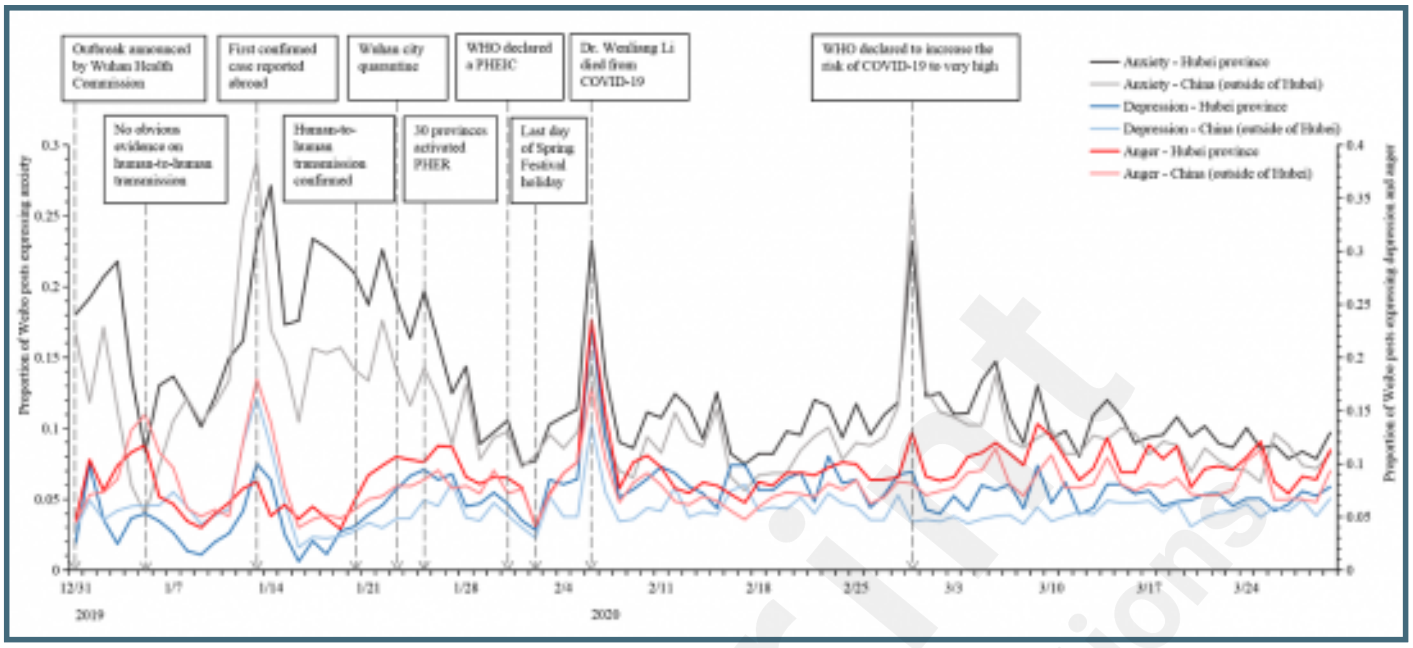
Daily number of COVID-19 related Weibo posts and confirmed COVID-19 cases in China.



Daily proportions of Weibo posts expressing risk perception.



Daily proportions of Weibo posts expressing anxiety, depression and anger.



Preprint JMIR Publications

Multimedia Appendixes

Preprint JMIR Publications

Ranking of key events in Weibo “Hot Search” during the COVID-19 epidemic.
URL: https://asset.jmir.pub/assets/ec218b97be5e7fe5caf9312ace719124.docx

Examples of keywords and Weibo posts assessing risk perception and mental health burden by the Linguistic Inquiry and Word Count program.
URL: https://asset.jmir.pub/assets/9405f4576206673ea6ca5ff5f9ff632d.docx

Preprint JMIR Publications